Learning rate experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why can equal epochs give different update counts when batch size changes?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Run a controlled comparison of learning rates. Try the recall before opening notes.

Your learning target

I can run a **controlled comparison** of **learning rate**s.

I can explain slow progress **oscillation** and increasing loss in a toy model.

I can report the limits of a **training** experiment.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Run a **controlled comparison** of **learning rate**s

Explain slow progress **oscillation** and increasing loss in a toy model

Report the limits of a **training** experiment

Retrieve what you already know

> Why can equal epochs give different update counts when batch size changes?

My starting explanation

This week's end goal

By the end, I can explain **learning rate** experiments using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Learning rate experiments case. Separate observations, given facts and assumptions before choosing an explanation.

Use the same restricted network h = ReLU(x), y = vh, with only v trained. Keep the fictional **training** pairs (1, 2) and (2, 4), start v = 0, and use one full-data batch per update.

Compare rates 0.05, 0.2, 0.3, 0.4 and 0.5 for exactly four updates each. Restart at zero for every run.

The **training** objective is ordinary MSE: L(v) = 2.5 times (v - 2) squared. Its gradient is 5 times (v - 2). A step is new v = v - rate times 5 times (v - 2).

This small quadratic lets us explain each result rather than treating the rate as a magic setting.

Rate	v after first update	First training MSE	v after four updates
0.05	0.5	5.625	1.3671875
0.2	2	0	2
0.3	3	2.5	1.875
0.4	4	10	0
0.5	5	22.5	-8.125

The same rate need not behave this way for another model, **input** scale, loss convention or batch rule. Rate 0.2 is an exact one-step solution here because of this particular gradient formula.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Learning rate experiments. Connect each labelled input, deciding step and result.

Write d = v - 2, the signed distance from this objective minimum. Substituting the update gives new d = (1 - 5 times rate) times old d. At rate 0.05 the factor is 0.75: distance shrinks without changing sign.

At rate 0.3 it is -0.5: distance changes sign and halves. Both can reduce loss.

```
def run(rate, updates=4):
    v = 0.0
    history = [v]
    for _ in range(updates):
        gradient = 5 * (v - 2)
        v = v - rate * gradient
        history.append(v)
    losses = [2.5 * (value - 2) ** 2 for value in history]
    return history, losses

for rate in [0.05, 0.2, 0.3, 0.4, 0.5]:
    values, losses = run(rate)
    print(rate, values, losses)
```

The helper assumes a finite rate and nonnegative integer update count. Its five logged states include the initial value.

At 0.3 the parameter sequence is [0, 3, 1.5, 2.25, 1.875] and loss is [10, 2.5, 0.625, 0.15625, 0.0390625]. Crossing the minimum does not itself mean **training** failed.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

Compare the supplied cases for Learning rate experiments. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Add a rate 1 run from a = 0.

Predicted result

Prediction: Its path is 8 then 0, with loss 16 at both steps.

Why the result follows

Reason: The update alternates across 4 without reducing the initial loss.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Learning rate experiments to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Learning rate	The multiplier controlling the size of a gradient update
Controlled comparison	An experiment changing a named factor while holding other stated conditions fixed
Oscillation	Repeated movement from one side of a value to the other
Divergence	Movement away from a target solution with growing error in this experiment

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Learning rate experiments, predict a step, then compare with its explanation.

The update factor explains three behaviours

RATE AND FACTOR	SIGNED DISTANCE BEHAVIOUR	TRAINING LOSS BEHAVIOUR
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

For rate 0.4, calculate its distance multiplier, the first two new v values and both losses. Does a changing parameter guarantee progress?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learning rate experiments

Compare **learning rate**s with identical starting loss and update counts.

The facts we will use

Supplied facts

Loss $J(a)=(a-4)^2$; gradient $2(a-4)$; start $a = 0$. Compare two updates at rates 0.25 and 0.75.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/14

Learning rate experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Rate 0.25

First a = 2, loss 4; second a = 3, loss 1.



2. Rate 0.75

First a = 6, loss 4; second a = 3, loss 1.



3. Inspect paths

Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

CHANGE ONE THING

Add a rate 1 run from a = 0.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Rate 0.25	First a = 2, loss 4; second a = 3, loss 1.
2. Rate 0.75	First a = 6, loss 4; second a = 3, loss 1.
3. Inspect paths	Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

What this example does not establish

Keep the stated limit

These quadratic paths explain step size. Real nonconvex networks can have more

complicated rate behaviour.

Keep the stated limit

Read the labelled picture: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Add a rate 1 run from a = 0. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Learning rate experiments claim and explain a repair using the supplied evidence.

Check the claim against the facts

For a nonzero starting distance, loss strictly decreases on each step of this quadratic when the absolute value of 1 - 5 times rate is below 1. For positive rates this means 0 < rate < 0.4. At 0 the model does not move.

At 0.4 it oscillates with unchanged loss. Above 0.4 its distance grows. These bounds belong to this model and objective, not to neural networks generally.

Check the claim against the facts

If we switch from mean squared error to the sum over the same two examples, the gradient doubles from 5(v - 2) to 10(v - 2). Keeping the old rate changes the update. To reproduce the original steps, halve the rate.

Likewise, rescaling features can change gradient magnitudes. Record the loss and units alongside the rate.

Locate the earliest unsupported step

Replace MSE by the sum of squared errors on the two **training** pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Learning rate experiments with support, then explain the deciding step in your own words.

For rate 0.05, start v = 0 and compute two updates and their **training** MSE values using the lesson objective.

For rate 0.4, calculate its distance multiplier, the first two new v values and both losses. Does a changing parameter guarantee progress?

Explain your prediction to a partner before comparing results.

Practise with less help: Explain a crossing

Use workbook W2 for the connected example "**Learning rate** experiments". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Learning rate** experiments. Proposed explanation: "Crossing the minimum means **divergence**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Learning rate experiments. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

At rate 0.3 the first update gives v = 3, past the minimum at 2. Calculate the next v and loss. Has the first crossing proved **divergence**?

```
def run(rate, updates=4):
    v = 0.0
    history = [v]
    for _ in range(updates):
        gradient = 5 * (v - 2)
        v = v - rate * gradient
        history.append(v)
    losses = [2.5 * (value - 2) ** 2 for value in history]
    return history, losses

for rate in [0.05, 0.2, 0.3, 0.4, 0.5]:
    values, losses = run(rate)
    print(rate, values, losses)
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Learning rate experiments. Explain why each step is justified.

Explain → apply → challenge

For rate 0.05, start v = 0 and compute two updates and their **training** MSE values using the lesson objective.

Replace MSE by the sum of squared errors on the two **training** pairs. Which rate reproduces the MSE run at rate 0.3? Explain using gradients.

Run A starts v = 0 with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final **training** losses. Can the difference be attributed only to rate?

Propose a repair.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **learning rate** experiments **evidence** sheet to Image-pattern Network Investigation. Show how you can run a **controlled comparison** of **learning rate**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Record the initial state, data, frozen values, objective, batch rule, update budget, candidate rates, parameter traces and loss traces. These runs compare **training** behaviour on two examples.

They do not establish which setting performs best on independent data. Keep the final test set out of rate selection; later lessons use validation curves to make development choices.

> **A team says rate 0.2 is the best rate for every neural network because it reached zero training loss here. Replace this claim with a precise experimental conclusion and one limitation.**

Next week compares **training** and validation curves, including when their trends disagree.

Run A starts v = 0 with full-data batches for four updates. Run B starts at the final A value, changes both rate and batch size, then compares final **training** losses. Can the difference be attributed only to rate?

Propose a repair.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Learning rate experiments case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

The update factor explains three behaviours

RATE AND FACTOR	SIGNED DISTANCE BEHAVIOUR	TRAINING LOSS BEHAVIOUR
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Compare a new objective independently

Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use J(a) = (a - 1) squared with gradient 2(a - 1), start a = 0 and two updates. Restart for each rate 0.25, 0.5 and 1. Compute the two new a values and losses for each run. Identify the result supported after two updates.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Learning rate experiments, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Compare **learning rate**s with identical starting loss and update counts.

By the end, I can	How I will show it
Run a controlled comparison of learning rates	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Explain slow progress oscillation and increasing loss in a toy model	Explain the deciding step, show a trace or test, and state one limit.
Report the limits of a training experiment	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Learning rate experiments without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Run a **controlled comparison** of **learning rate**s

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare **learning rate**s with identical starting loss and update counts.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

1. Inspect paths

2. Rate 0.25

3. Rate 0.75

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect paths Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add a rate 1 run from a = 0. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Its path is 8 then 0, with loss 16 at both steps.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its path is 8 then 0, with loss 16 at both steps. The update alternates across 4 without reducing the initial loss.

Repair a missing building block

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Updating parameters (ai-5/11). Goal: Apply a **learning rate** to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

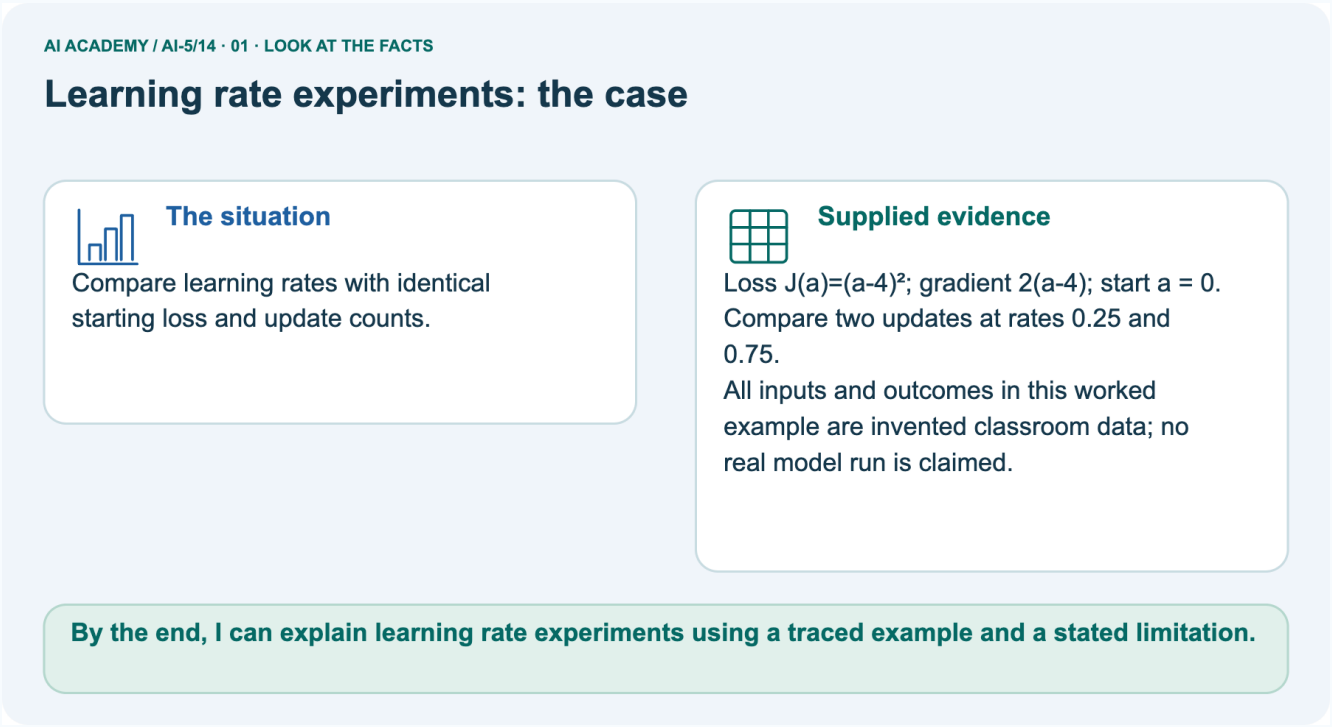
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Learning rate experiments graphic using its labels, deciding evidence and relevant limitation.

01. Learning rate experiments: the case



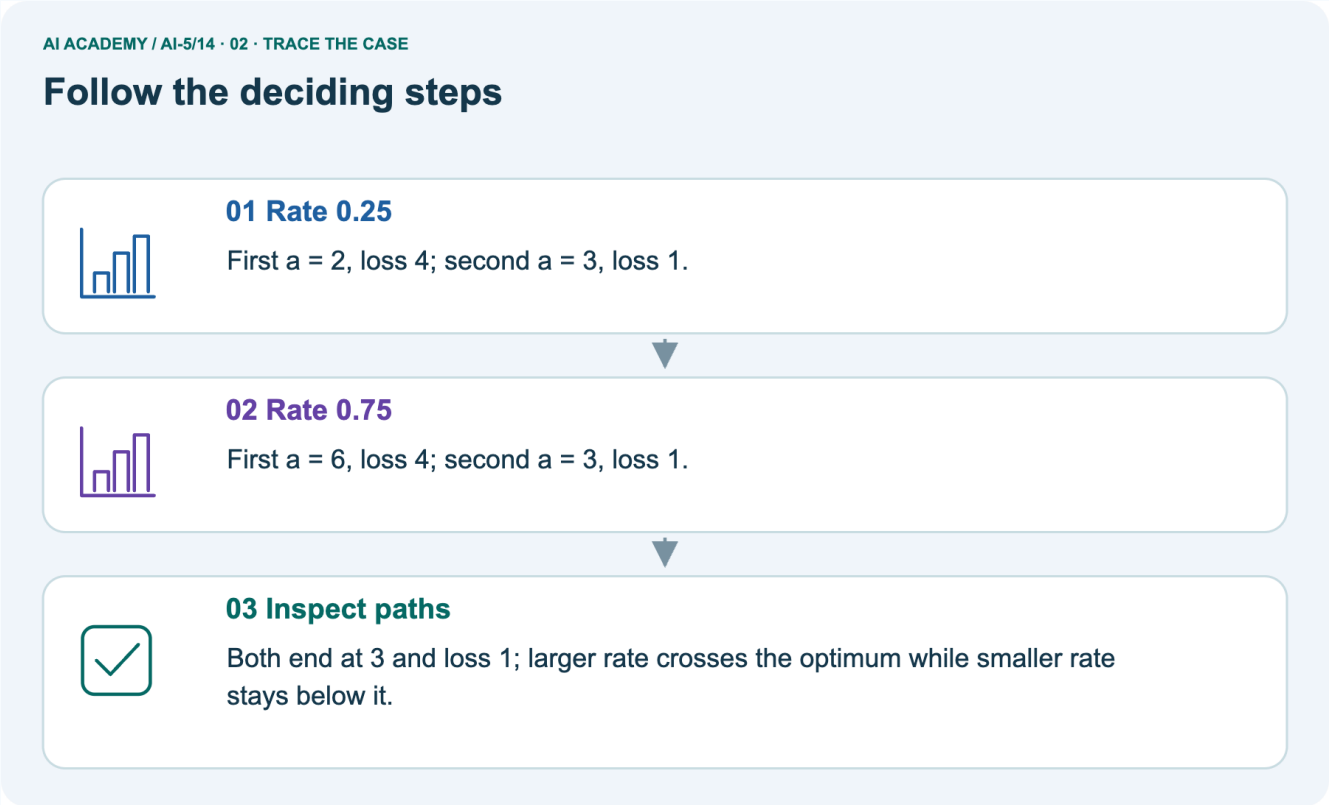
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare **learning rate**s with identical starting loss and update counts. Supplied facts: Loss J(a)=(a-4)²; gradient 2(a-4); start a = 0. Compare two updates at rates 0.25 and 0.75.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Rate 0.25: First a = 2, loss 4; second a = 3, loss 1. → Rate 0.75: First a = 6, loss 4; second a = 3, loss 1. → Inspect paths: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it.

> Point to each numbered stage and explain which recorded **input** it uses.

03. The update factor explains three behaviours

AI ACADEMY / AI-5/14 · 03 · INSPECT THE LESSON MODEL

The update factor explains three behaviours

Rate and factor	Signed distance behaviour	Training loss behaviour
0.05 gives 0.75	Same side and shrinking	Multiplied by 0.5625
0.3 gives -0.5	Alternating and shrinking	Multiplied by 0.25
0.5 gives -1.5	Alternating and growing	Multiplied by 2.25

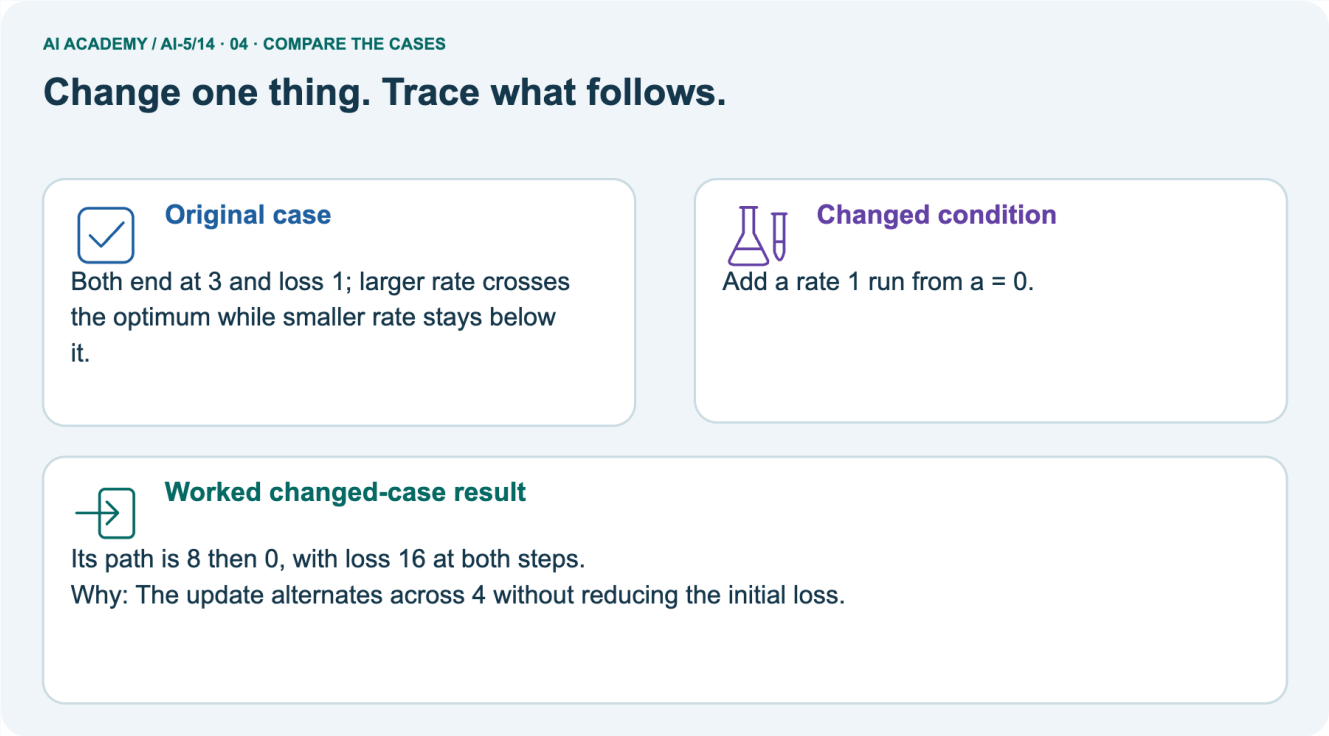
For this objective, loss is 2.5 times squared distance. Therefore the loss multiplier is the square of the distance multiplier.

Follow the labelled picture with the matching explanation. Open the vector diagram

The update factor explains three behaviours. 0.05 gives 0.75 | Same side and shrinking | Multiplied by 0.5625. 0.3 gives -0.5 | Alternating and shrinking | Multiplied by 0.25. 0.5 gives -1.5 | Alternating and growing | Multiplied by 2.25

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



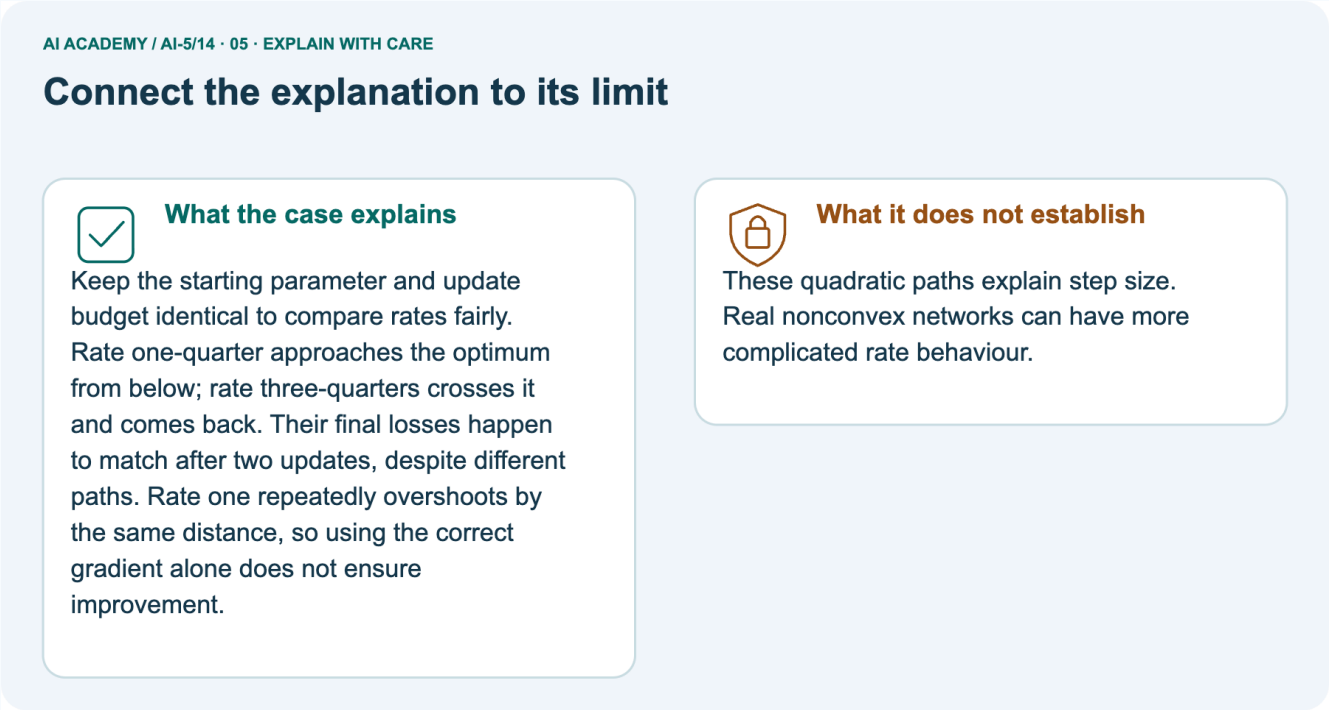
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both end at 3 and loss 1; larger rate crosses the optimum while smaller rate stays below it. Changed condition: Add a rate 1 run from a = 0. Expected result: Its path is 8 then 0, with loss 16 at both steps.

Reason: The update alternates across 4 without reducing the initial loss.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

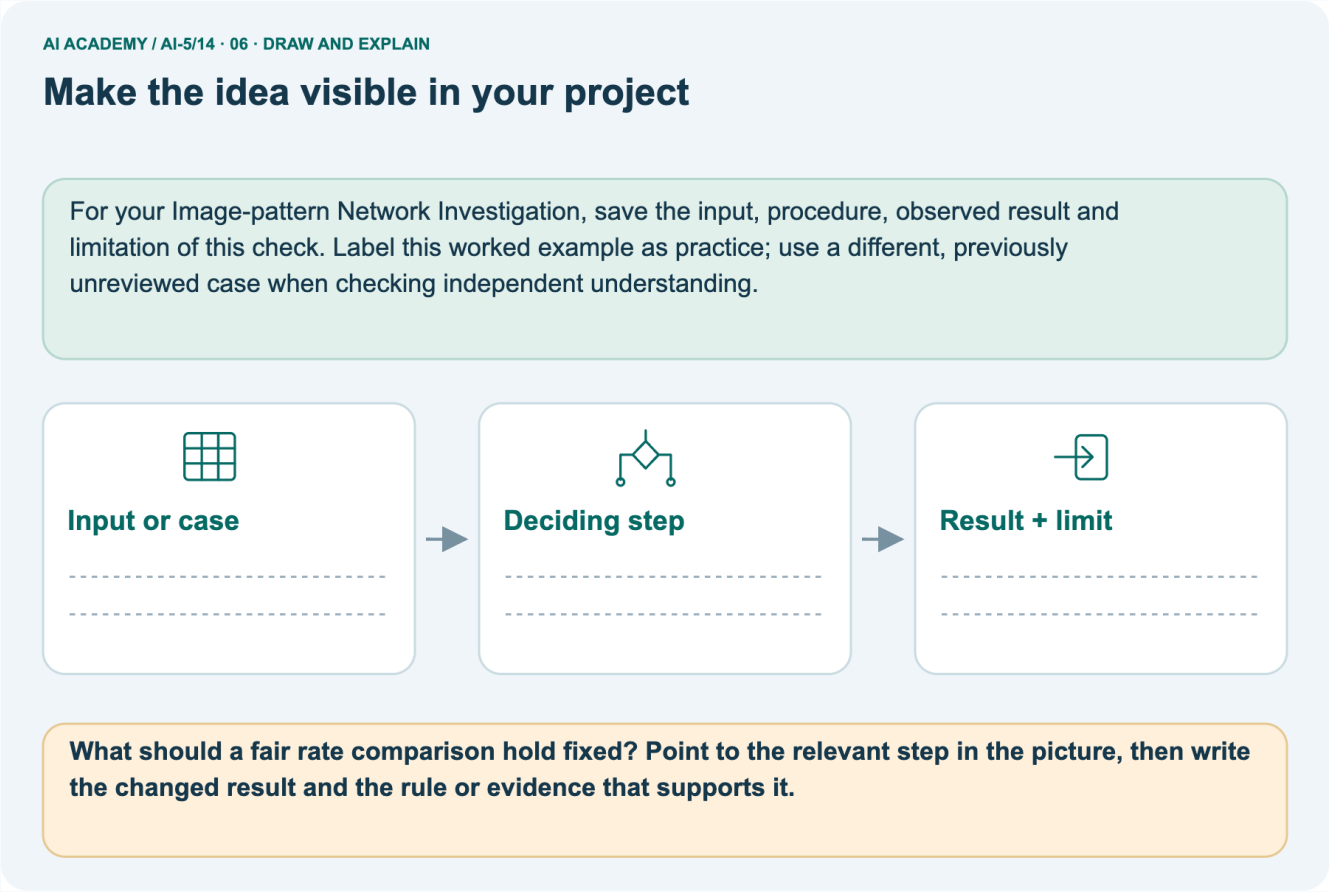
Keep the starting parameter and update budget identical to compare rates fairly. Rate one-quarter approaches the optimum from below; rate three-quarters crosses it and comes back.

Their final losses happen to match after two updates, despite different paths. Rate one repeatedly overshoots by the same distance, so using the correct gradient alone does not ensure improvement.

Limit: These quadratic paths explain step size. Real nonconvex networks can have more complicated rate behaviour.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.